Digital Logic Design

Chapter 1

Digital Systems and Binary Numbers

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Outline of Chapter 1

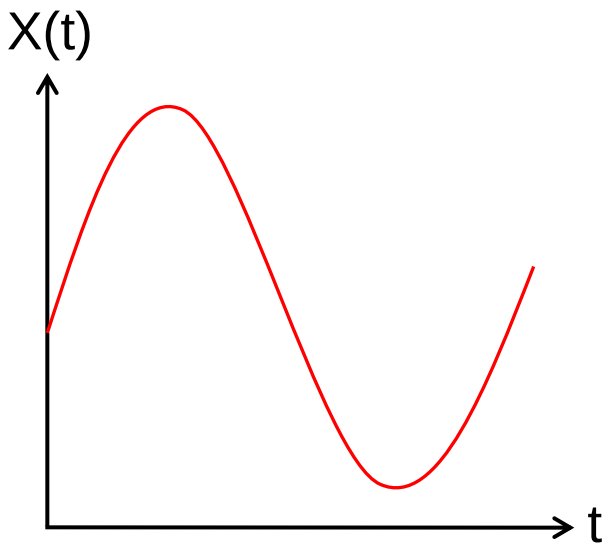
- Digital Systems
- Binary Numbers
- Number-base Conversions
- Octal and Hexadecimal Numbers
- Complements
- Signed Binary Numbers
- Binary Codes
- Binary Storage and Registers
- Binary Logic

Digital Systems and Binary Numbers

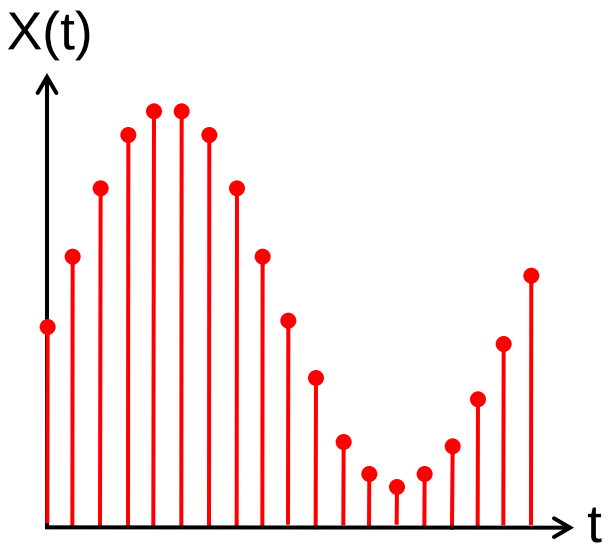
- Digital age and information age
- Digital computers
  - General purposes
  - Many scientific, industrial and commercial applications
- Digital systems
  - Telephone switching exchanges
  - Digital camera
  - Electronic calculators, PDA's
  - Digital TV
- Discrete information-processing systems
  - Manipulate discrete elements of information
  - For example, {1, 2, 3, …} and {A, B, C, …}…

Analog and Digital Signal

- Analog system
  - The physical quantities or signals may vary continuously over a specified range.
- Digital system
  - The physical quantities or signals can assume only discrete values.
  - Greater accuracy



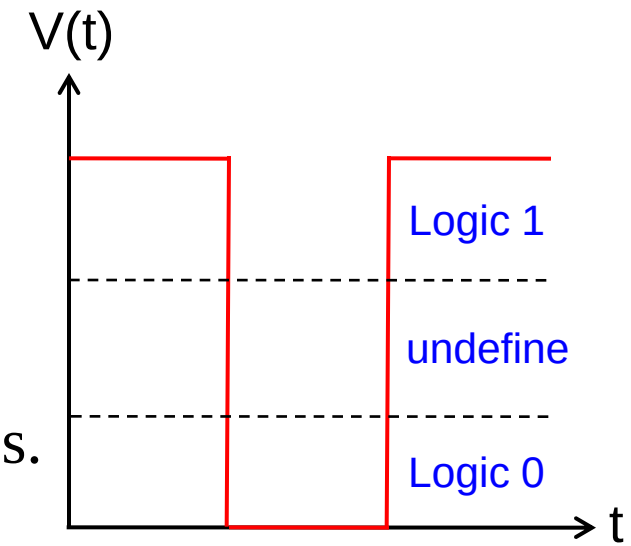
Analog signal



Digital signal

Binary Digital Signal

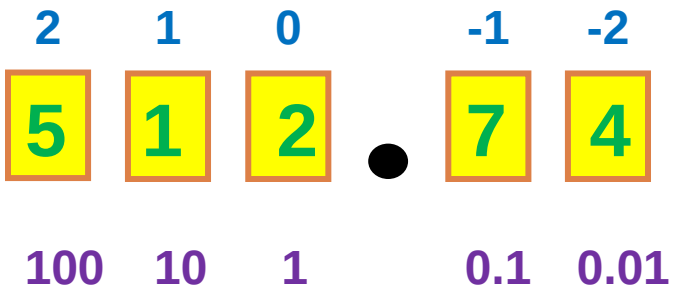
- An information variable represented by physical quantity.
- For digital systems, the variable takes on discrete values.
  - Two level, or binary values are the most prevalent values.
- Binary values are represented abstractly by:
  - Digits 0 and 1
  - Words (symbols) False (F) and True (T)
  - Words (symbols) Low (L) and High (H)
  - Words On and Off
- Binary values are represented by values or ranges of values of physical quantities.



Binary digital signal

Decimal Number System

- Base (also called radix) = 10
  - 10 digits { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }
- Digit Position
  - Integer & fraction
- Digit Weight
  - Weight = (*Base*) ***Position***
- Magnitude
  - Sum of "***Digit*** x *Weight*"
- Formal Notation



$$= d_2 * B^2 + d_1 * B^1 + d_0 * B^0 + d_{-1} * B^{-1} + d_{-2} * B^{-2}$$

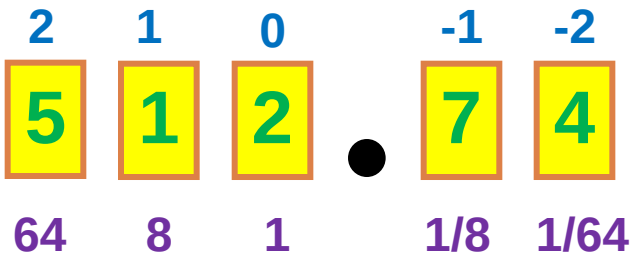
$$= 5*10^2 + 1*10^1 + 2*10^0 + 7*10^{-1} + 4*10^{-2}$$

$$= 5*100 + 1*10 + 2*1 + 7*0.1 + 4*0.01$$

$$= (512.74)_{10}$$

Octal Number System

- Base = 8
  - 8 digits { 0, 1, 2, 3, 4, 5, 6, 7 }
- Digit Position
  - Integer & fraction
- Digit Weight
  - Weight = (*Base*) ***Position***
- Magnitude
  - Sum of "***Digit*** x *Weight*"
- Formal Notation



$$= d_2 * B^2 + d_1 * B^1 + d_0 * B^0 + d_{-1} * B^{-1} + d_{-2} * B^{-2}$$

$$= 5*8^2 + 1*8^1 + 2*8^0 + 7*8^{-1} + 4*8^{-2}$$

$$= 5*64 + 1*8 + 2*1 + 7*\frac{1}{8} + 4*\frac{1}{64}$$

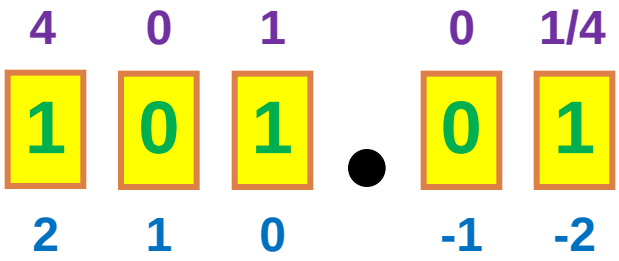
$$= (330.9375)_{10}$$

Binary Number System

- Base = 2
  - 2 digits { 0, 1 }, called *b*inary dig*its* or "*bits*"
- Digit Position
  - Integer & fraction
- Digit Weight
  - Weight = (*Base*) ***Position***
- Magnitude
  - Sum of "***Digit*** x *Weight*"
- Formal Notation
- Groups of bits

  4 bits = *Nibble*

  8 bits = *Byte*



$$= d_2 * B^2 + d_1 * B^1 + d_0 * B^0 + d_{-1} * B^{-1} + d_{-2} * B^{-2}$$

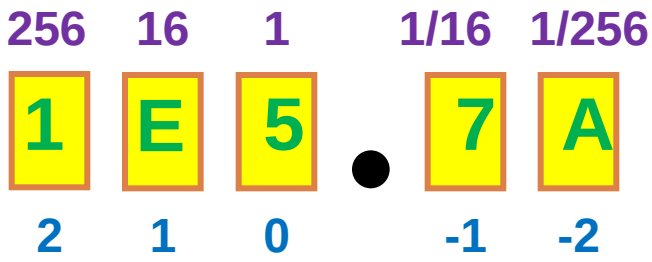
$$= 1*2^2 + 0*2^1 + 1*2^0 + 0*2^{-1} + 1*2^{-2}$$

$$= 1*4 + 0*2 + 1*1 + 0*\frac{1}{2} + 1*\frac{1}{4}$$

$$= (5.25)_{10}$$

Hexadecimal Number System

- Base = 16
  - 16 digits { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F }
- Digit Position
  - Integer & fraction
- Digit Weight
  - Weight = (*Base*) ***Position***
- Magnitude
  - Sum of "***Digit*** x *Weight*"
- Formal Notation



$$= d_2 * B^2 + d_1 * B^1 + d_0 * B^0 + d_{-1} * B^{-1} + d_{-2} * B^{-2}$$

$$= 1*16^2 + 14*16^1 + 5*16^0 + 7*16^{-1} + 10*16^{-2}$$

$$= 1*256 + 14*16 + 5*1 + 7*\frac{1}{16} + 10*\frac{1}{256}$$

$$= (485.4765625)_{10}$$

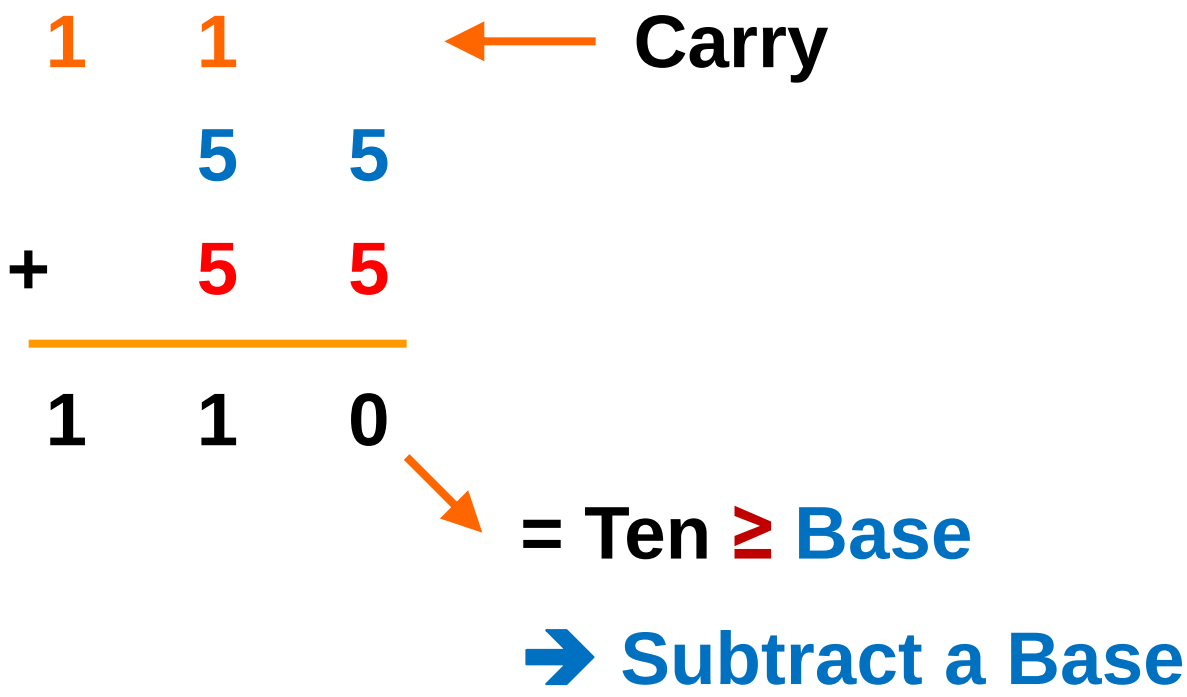
The Power of 2

n	2^n
0	$2^0=1$
1	$2^1=2$
2	$2^2=4$
3	$2^3=8$
4	$2^4=16$
5	$2^5=32$
6	$2^6=64$
7	$2^7=128$

n	2^n	
8	$2^8=256$	
9	$2^9=512$	
10	$2^{10}=1024$	Kilo
11	$2^{11}=2048$	
12	$2^{12}=4096$	
20	$2^{20}=1M$	Mega
30	$2^{30}=1G$	Giga
40	$2^{40}=1T$	Tera

Decimal Addition

- Decimal Addition



Binary Addition

- Column Addition

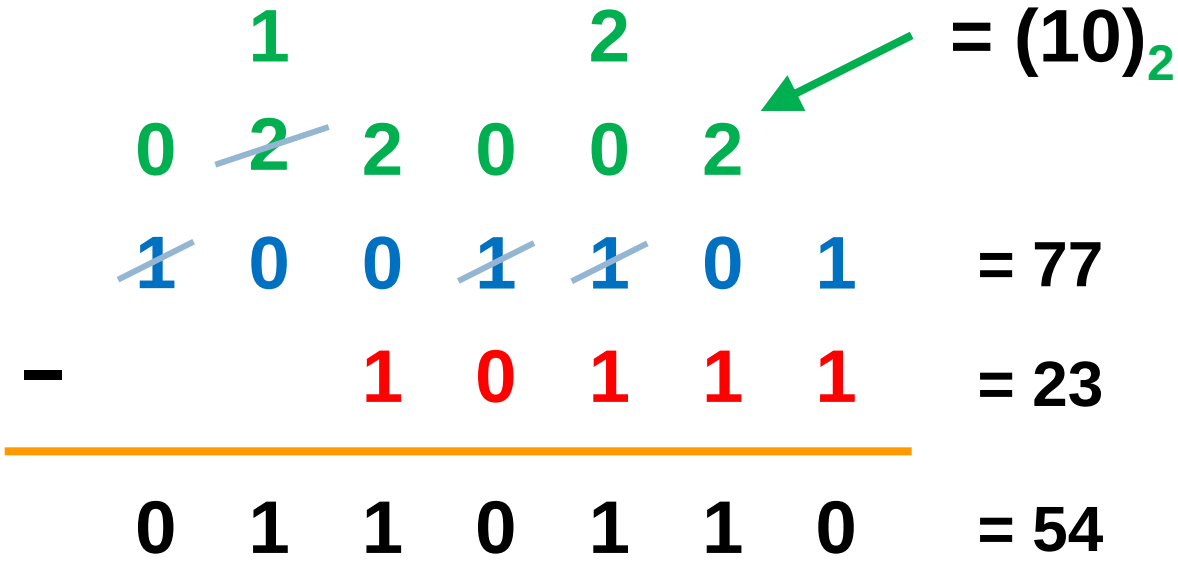
```
  1  1  1  1  1  1
     1  1  1  1  0  1     = 61
+       1  0  1  1  1     = 23
-----------------------
  1  0  1  0  1  0  0     = 84
```

= Two ≥ Base

➔ Subtract a Base

Binary Subtraction

- Borrow a "Base" when needed

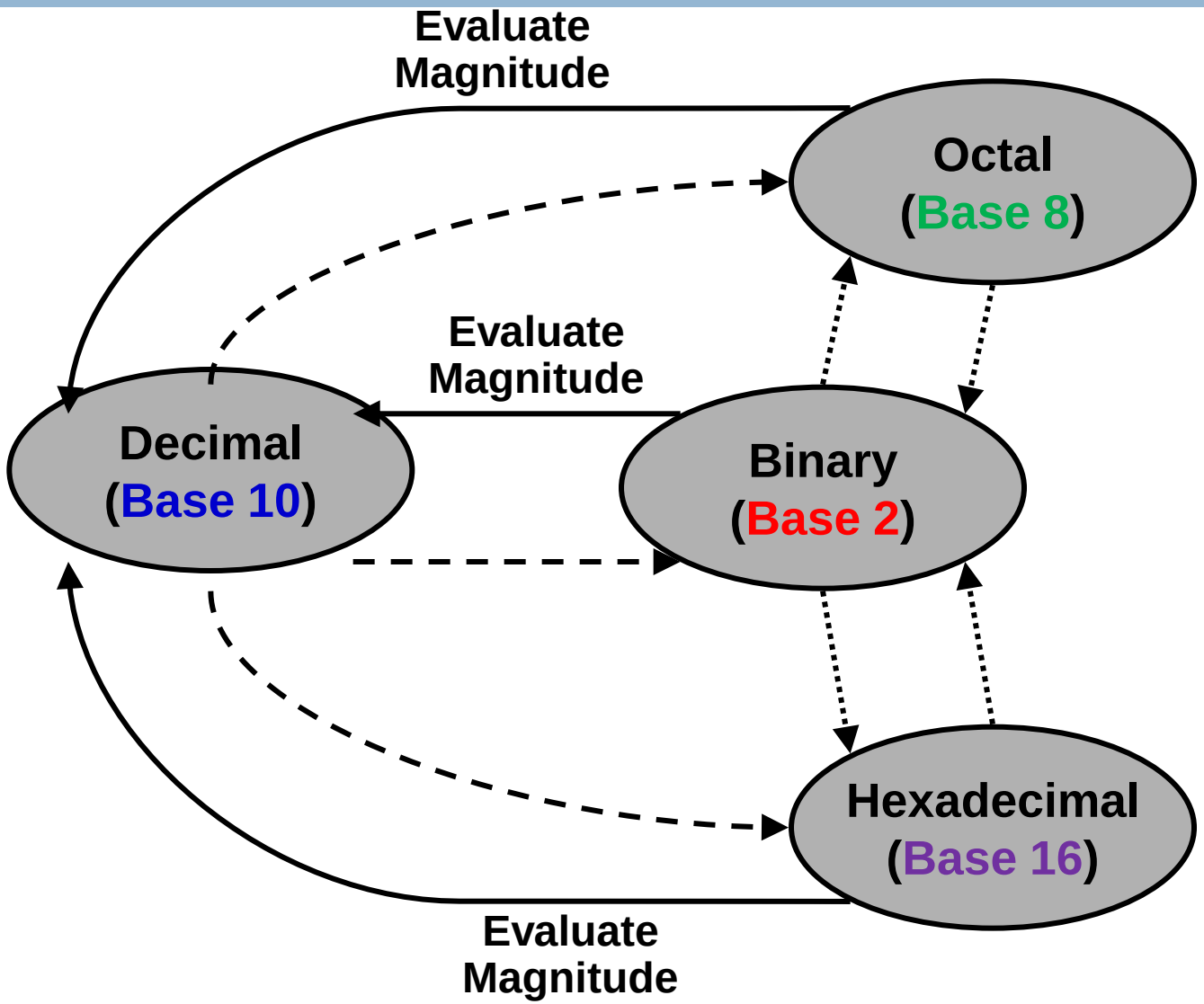


Binary Multiplication

- Bit by bit

			1	0	1	1	1
x				1	0	1	0
			0	0	0	0	0
		1	0	1	1	1	
	0	0	0	0	0		
1	0	1	1	1			
1	1	1	0	0	1	1	0

Number Base Conversions



Decimal (Integer) to Binary Conversion

- Divide the number by the 'Base' (=2)
- Take the remainder (either 0 or 1) as a coefficient
- Take the quotient and repeat the division

Example: $(13)_{10}$

	Quotient	Remainder	Coefficient
13 / 2 =	6	1	$a_0 = 1$
6 / 2 =	3	0	$a_1 = 0$
3 / 2 =	1	1	$a_2 = 1$
1 / 2 =	0	1	$a_3 = 1$

Answer: $(13)_{10} = (a_3\ a_2\ a_1\ a_0)_2 = (1101)_2$

MSB LSB

Decimal (*Fraction*) to Binary Conversion

- Multiply the number by the 'Base' (=2)
- Take the integer (either 0 or 1) as a coefficient
- Take the resultant fraction and repeat the division

Example: $(0.625)_{10}$

		Integer		Fraction	Coefficient
0.625	* 2 =	1	.	25	$a_{-1} = 1$
0.25	* 2 =	0	.	5	$a_{-2} = 0$
0.5	* 2 =	1	.	0	$a_{-3} = 1$

Answer: $(0.625)_{10} = (0.a_{-1}\, a_{-2}\, a_{-3})_2 = (0.101)_2$

MSB LSB

Decimal to Octal Conversion

Example: $(175)_{10}$

	Quotient	Remainder	Coefficient
175 / 8 =	21	7	$a_0 = 7$
21 / 8 =	2	5	$a_1 = 5$
2 / 8 =	0	2	$a_2 = 2$

Answer: $(175)_{10} = (a_2\, a_1\, a_0)_8 = (257)_8$

Example: $(0.3125)_{10}$

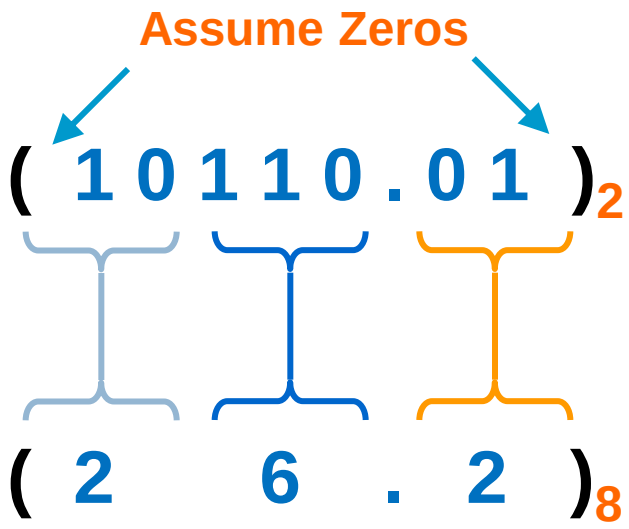
	Integer		Fraction	Coefficient
0.3125 * 8 =	2	.	5	$a_{-1} = 2$
0.5 * 8 =	4	.	0	$a_{-2} = 4$

Answer: $(0.3125)_{10} = (0.a_{-1}\, a_{-2}\, a_{-3})_8 = (0.24)_8$

Binary – Octal Conversion

- $8 = 2^3$
- Each group of 3 bits represents an octal digit

Example:



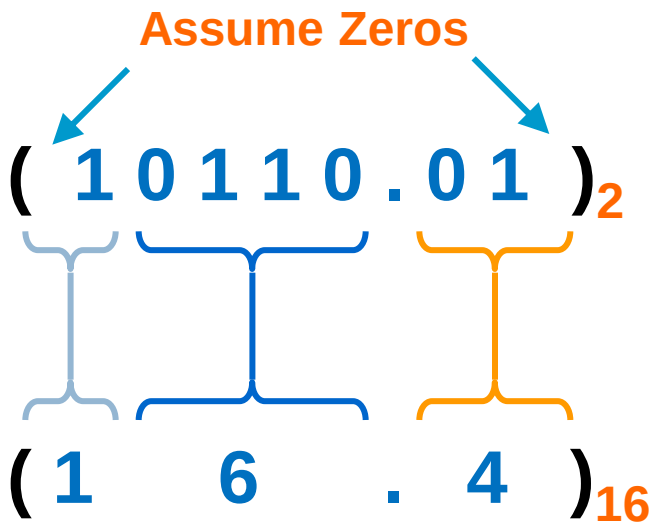
Octal	Binary
0	0 0 0
1	0 0 1
2	0 1 0
3	0 1 1
4	1 0 0
5	1 0 1
6	1 1 0
7	1 1 1

Works both ways (*Binary to Octal & Octal to Binary*)

Binary – Hexadecimal Conversion

- $16 = 2^4$
- Each group of 4 bits represents a hexadecimal digit

Example:



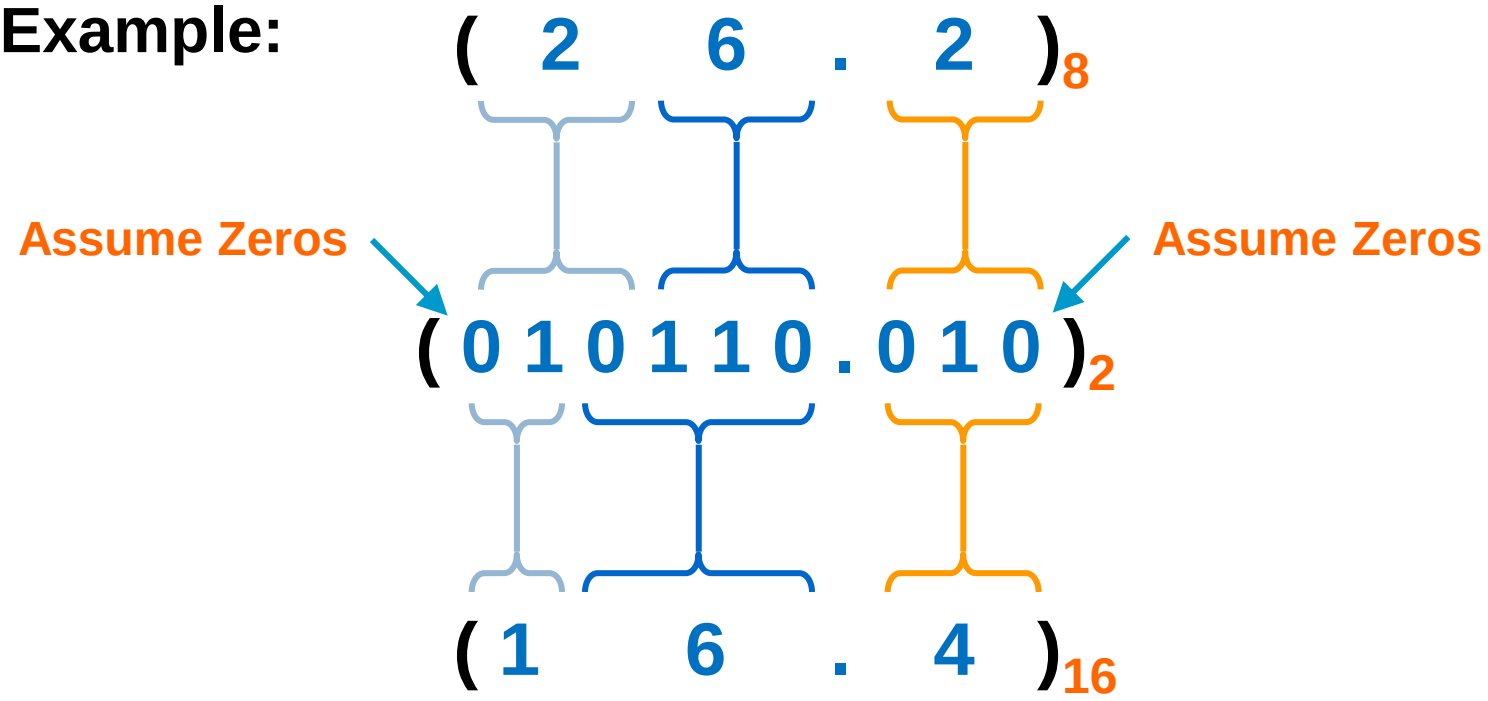
Hex	Binary
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1
A	1 0 1 0
B	1 0 1 1
C	1 1 0 0
D	1 1 0 1
E	1 1 1 0
F	1 1 1 1

Works both ways (*Binary to Hex & Hex to Binary*)

Octal – Hexadecimal Conversion

- Convert to Binary as an intermediate step

Example:



Works both ways (*Octal to Hex & Hex to Octal*)

Decimal, Binary, Octal and Hexadecimal

Decimal	Binary	Octal	Hex
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Complements

- There are two types of complements for each base-r system: the radix complement and diminished radix complement.
- Radix Complement: ***r's Complement***
  - Given a number *N* in base *r* having *n* digits, the *r*'s complement *of N* is defined as:

$$r^n - N$$

- Diminished Radix Complement: ***(r-1)'s Complement***

$$(r^n - 1) - N$$

Radix Complements

The *r's* complement of an n-digit number *N* in base *r* is defined as $r^n - N$ for N ≠ 0 and as 0 for N = 0. Comparing with the (r – 1) 's complement, we note that the r's complement is obtained by adding 1 to the (r – 1) 's complement, since $r^n - N = [(r^n - 1) - N] + 1$.

The 10's complement of 012398 is 987602
The 10's complement of 246700 is 753300

$r^n - N$
$10^6 - 012398$

9	9	9	9	9	10
0	1	2	3	9	8
9	8	7	6	0	2

The 2's complement of 1101100 is 0010100
The 2's complement of 0110111 is 1001001

$r^n - N$
$2^7 - 1101100$

1	1	1	1	2	0	0
1	1	0	1	1	0	0
0	0	1	0	1	0	0

Radix Complements

- 2's Complement of Base-2

OR

- Take 1's complement then add 1
- Toggle all bits to the left of the first '1' from the right

***Example*:**

Number: **1 0 1 1 0 0 0 0**

1's Complement: **0 1 0 0 1 1 1 1**

$$
\begin{array}{rr}
 & 1\,0\,1\,1\,0\,0\,0\,0 \\
 & 0\,1\,0\,0\,1\,1\,1\,1 \\
+ & \qquad\qquad\quad 1 \\
\hline
 & 0\,1\,0\,1\,0\,0\,0\,0
\end{array}
$$

Diminished Radix Complements

- **Example for 6-digit decimal numbers:**
  - 9's complement is $(r^n - 1) - N = (10^6 - 1) - N = 999999 - N$
  - 9's complement of 546700 is 999999–546700 = 453299
- **Example for 7-digit binary numbers:**
  - 1's complement is $(r^n - 1) - N = (2^7 - 1) - N = 1111111 - N$
  - 1's complement of 1011000 is 1111111–1011000 = 0100111
- **Observation:**
  - Subtraction from $(r^n - 1)$ will never require a borrow
  - Diminished radix complement can be computed digit-by-digit
  - For binary: 1 – 0 = 1 and 1 – 1 = 0

Diminished Radix Complements

- 1's Complement of Base-2
  - All '0's become '1's
  - All '1's become '0's

Example $(10110000)_2$

$\Rightarrow (01001111)_2$

If you add a number and its 1's complement …

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0 \\ +\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 1 \\ \hline 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1 \end{array}$$

Subtraction with Complements

- Subtraction with Complements
  - The subtraction of two *n*-digit unsigned numbers $M - N$ in base r can be done as follows:

1. Add the minuend M to the r's complement of the subtrahend N. Mathematically, $M + (r^n - N) = M - N + r^n$.
2. If $M \geqq N$, the sum will produce and end carry r^n, which can be discarded; what is left is the result $M - N$.
3. If $M < N$, the sum does not produce an end carry and is equal to $r^n - (N - M)$, which is the r's complement of $(N - M)$. To obtain the answer in a familiar form, take the r's complement of the sum and place a negative sign in front.

Subtraction with Complements

- Example
  - Using 10's complement, subtract 72532 – 3250.

	$M =$	72532
10's complement of	$N =$	+96750
	Sum =	169282
	Discard end carry 10^5 =	–100000
	Answer =	69282

- Example 1.6
  - Using 10's complement, subtract 3250 – 72532.

	M =	03250
10's complement of	N =	+27468
	Sum =	30718

There is no end carry.

Therefore, the answer is – (10's complement of 30718) = – 69282.

Subtraction with Complements

- Example
  - Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$; and (b) $Y - X$, by using 2's complement.

(a)		
	X =	1010100
	2's complement of Y =	+0111101
	Sum =	10010001
	Discard end carry 2^7 =	−10000000
	Answer. X – Y =	0010001

(b)		
	Y =	1000011
	2's complement of X =	+ 0101100
	Sum =	1101111

There is no end carry. Therefore, the answer is Y – X = – (2's complement of 1101111) = – 0010001.

Subtraction with Complements

- Subtraction of unsigned numbers can also be done by means of the $(r-1)$'s complement. Remember that the $(r-1)$'s complement is one less then the r's complement.
- **Example**
  - Repeat Example 1.7, but this time using 1's complement.

(a) $X - Y = 1010100 - 1000011$

$X =$	1010100
1's complement of $Y =$	± 0111100
Sum =	10010000
End-around carry =	+ 1
Answer. $X - Y =$	0010001

(b) $Y - X = 1000011 - 1010100$

$Y =$	1000011
1's complement of $X =$	+ 0101011
Sum =	1101110

There is no end carry, Therefore, the answer is Y – X = – (1's complement of 1101110) = – 0010001.

Signed Binary Numbers

- To represent negative integers, we need a notation for negative values.
- In binary digits, it is customary to represent the sign with a bit placed in the leftmost position of the number.
- The convention is to make the sign bit 0 for positive and 1 for negative.
- **Example:** 8-bit signed binary numbers (-9)

Signed-magnitude representation:	10001001
Signed-1's-complement representation:	11110110
Signed-2's-complement representation:	11110111

- Table 1.3 lists all possible 4-bit signed binary numbers in the three representations.

Signed Binary Numbers

Table 1.3
Signed Binary Numbers

Decimal	Signed-2's Complement	Signed-1's Complement	Signed Magnitude
+7	0111	0111	0111
+6	0110	0110	0110
+5	0101	0101	0101
+4	0100	0100	0100
+3	0011	0011	0011
+2	0010	0010	0010
+1	0001	0001	0001
+0	0000	0000	0000
−0	—	1111	1000
−1	1111	1110	1001
−2	1110	1101	1010
−3	1101	1100	1011
−4	1100	1011	1100
−5	1011	1010	1101
−6	1010	1001	1110
−7	1001	1000	1111
−8	1000	—	—

Signed Binary Numbers

- Arithmetic addition
  - The addition of two numbers in the signed-magnitude system follows the rules of ordinary arithmetic.
    - If the signs are the same: we add the two magnitudes and give the sum (given the common sign).
    - If the signs are different: we subtract the smaller magnitude from the larger (give the sign to the larger magnitude).
  - The addition of two signed binary numbers with negative numbers represented in signed-2's-complement form is obtained from the addition of the two numbers, including their sign bits.
  - A carry out of the sign-bit position is discarded.

Signed Binary Numbers

- Arithmetic addition using the signed-2's-complement form.
  - **Example:** Addition using the signed-2's-complement form, 8-bit signed binary numbers, 7bits plus the signed bit.
    - +6 + (+13)

+ 6	00000110	
+13	00001101	
+ 19	00010011	

    - -6 + (+13)

− 6	11111010	
+13	00001101	
+ 7	00000111	

    - +6 + (-13)

+ 6	00000110	
−13	11110011	
− 7	11111001	

    - -6 + (-13)

− 6	11111010	
−13	11110011	
− 19	11101101	

Signed Binary Numbers

- Arithmetic Subtraction
  - **In 2's-complement form:**

1. Take the 2's complement of the subtrahend (including the sign bit) and add it to the minuend (including sign bit).
2. A carry out of sign-bit position is discarded.

$$(\pm A)-(+B)=(\pm A)+(-B)$$
$$(\pm A)-(-B)=(\pm A)+(+B)$$

- **Example:**

(– 6) – (– 13) → (11111010 – 11110011)

→ (11111010 + 00001101)

→ 00000111 ---→ (+ 7)

Binary-Coded Decimal (BCD)

- BCD Code
  - A number with k decimal digits will require 4k bits in BCD.
  - Decimal 396 is represented in BCD with 12bits as 0011 1001 0110, with each group of 4 bits representing one decimal digit.
  - A decimal number in BCD is the same as its equivalent binary number only when the number is between 0 and 9.
  - The binary combinations 1010 through 1111 are not used and have no meaning in BCD.

Table 1.4
Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Binary-Coded Decimal (BCD)

- Example:
  - Consider decimal 185 and its corresponding value in BCD and binary:

$$(185)_{10} = (0001\ 1000\ 0101)_{BCD} = (10111001)_2$$

- BCD addition

4	0100	4	0100	8	1000
+5	+0101	+8	+1000	+9	+1001
9	1001	12	1100	17	10001
			+0110		+0110
			10010		10111

Binary-Coded Decimal (BCD)

- Example:
  - Consider the addition of 184 + 576 = 760 in BCD:

BCD	1	1		
	0001	1000	0100	184
	+0101	0111	0110	+576
Binary sum	0111	10000	1010	
Add 6		0110	0110	
BCD sum	0111	0110	0000	760

  - Decimal Arithmetic: (+375) + (-240) = +135

Hint: using 10's complement of BCD

0	375
+9	760
0	135

Binary Codes

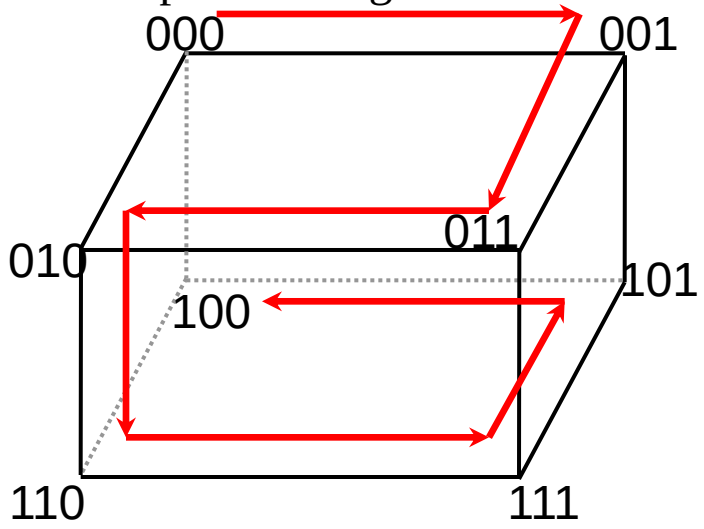
- Other Decimal Codes

Table 1.5
Four Different Binary Codes for the Decimal Digits

Decimal Digit	BCD 8421	2421	Excess-3	8, 4, −2, −1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111
	1010	0101	0000	0001
Unused	1011	0110	0001	0010
bit	1100	0111	0010	0011
combi-	1101	1000	1101	1100
nations	1110	1001	1110	1101
	1111	1010	1111	1110

Gray Codes

- Gray Code
  - The advantage is that only bit in the code group changes in going from one number to the next.
    - Error detection.
    - Representation of analog data.
    - Low power design.

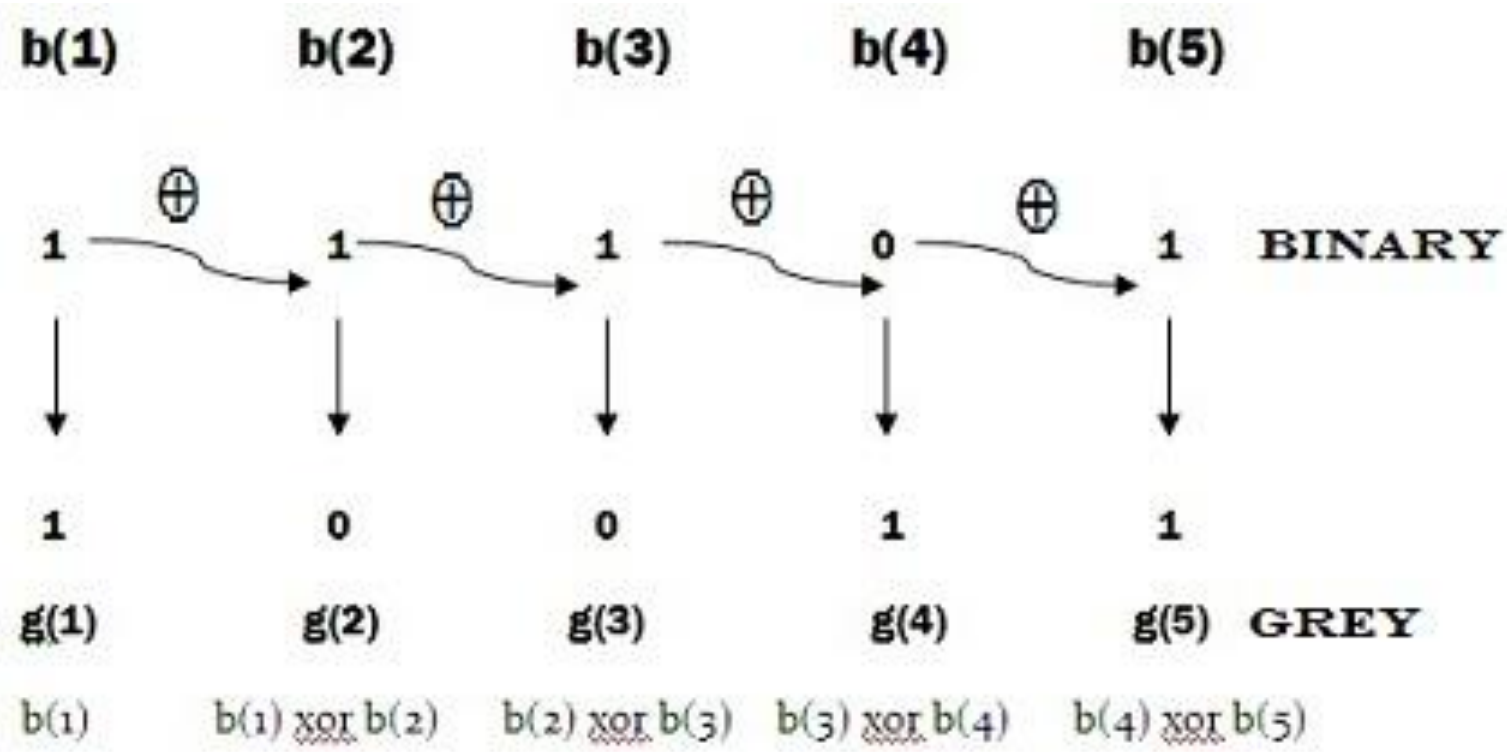


1-1 and onto!!

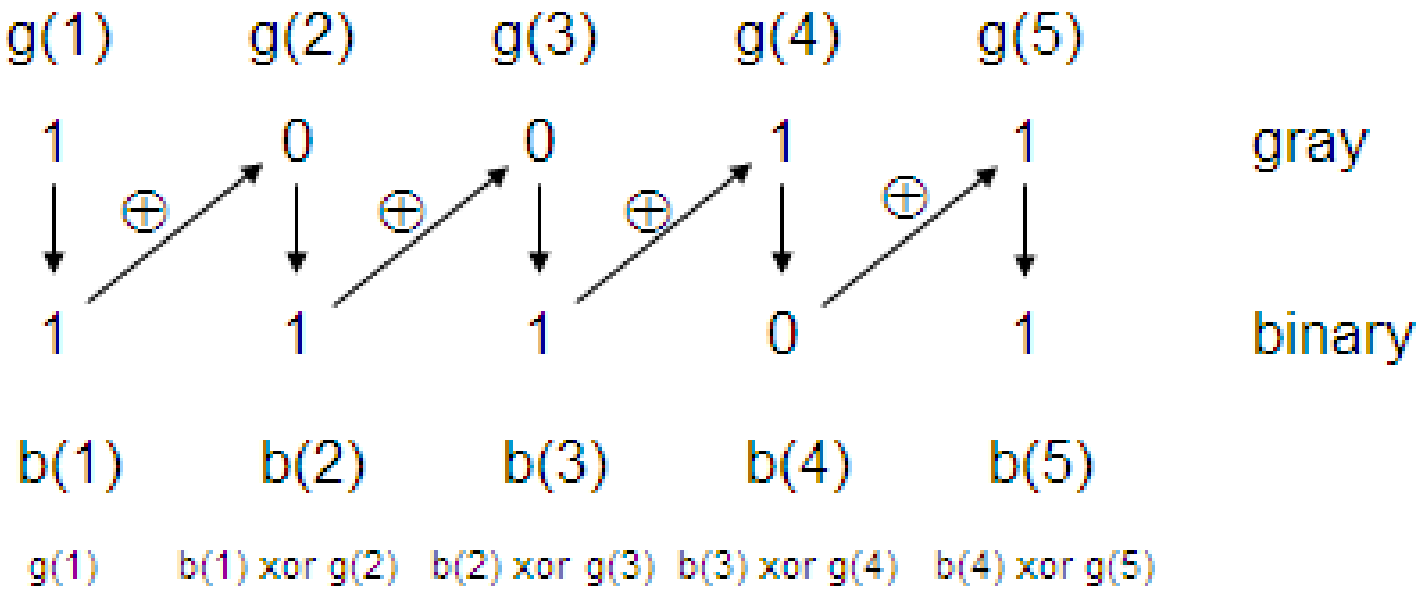
Table 1.6
Gray Code

Gray Code	Decimal Equivalent
0000	0
0001	1
0011	2
0010	3
0110	4
0111	5
0101	6
0100	7
1100	8
1101	9
1111	10
1110	11
1010	12
1011	13
1001	14
1000	15

Gray Codes


b(1) b(2) b(3) b(4) b(5)
1 1 1 0 1 BINARY
1 0 0 1 1
g(1) g(2) g(3) g(4) g(5) GREY
b(1) b(1) xor b(2) b(2) xor b(3) b(3) xor b(4) b(4) xor b(5)

Gray Codes


g(1) g(2) g(3) g(4) g(5)
1 0 0 1 1
gray
1 1 1 0 1
binary
b(1) b(2) b(3) b(4) b(5)
g(1) b(1) xor g(2) b(2) xor g(3) b(3) xor g(4) b(4) xor g(5)

ASCII Character Codes

- American Standard Code for Information Interchange (ASCII) is used to represent information sent as character-based data.
- It uses 7-bits to represent 128 characters:
  - 94 Graphic printing characters.
  - 34 Non-printing characters.
    - Some non-printing characters are used for text format (e.g. BS = Backspace, CR = carriage return).
    - Other non-printing characters are used for record marking and flow control (e.g. STX and ETX start and end text areas).

ASCII Properties

- ASCII has some interesting properties:
  - Digits 0 to 9 span Hexadecimal values 30_{16} to 39_{16}
  - Upper case A-Z span 41_{16} to $5A_{16}$
  - Lower case a-z span 61_{16} to $7A_{16}$
    - Lower to upper case translation (and vice versa) occurs by flipping bit 6.

ASCII Codes

- American Standard Code for Information Interchange (ASCII):

Table 1.7
American Standard Code for Information Interchange (ASCII)

	$b_7b_6b_5$							
$b_4b_3b_2b_1$	000	001	010	011	100	101	110	111
0000	NUL	DLE	SP	0	@	P	`	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	"	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	&	6	F	V	f	v
0111	BEL	ETB	'	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	,	<	L	\	l	|
1101	CR	GS	−	=	M	]	m	}
1110	SO	RS	.	>	N	^	n	~
1111	SI	US	/	?	O	−	o	DEL

ASCII Character Codes

- ASCII Character Code

Control characters

NUL	Null	DLE	Data-link escape
SOH	Start of heading	DC1	Device control 1
STX	Start of text	DC2	Device control 2
ETX	End of text	DC3	Device control 3
EOT	End of transmission	DC4	Device control 4
ENQ	Enquiry	NAK	Negative acknowledge
ACK	Acknowledge	SYN	Synchronous idle
BEL	Bell	ETB	End-of-transmission block
BS	Backspace	CAN	Cancel
HT	Horizontal tab	EM	End of medium
LF	Line feed	SUB	Substitute
VT	Vertical tab	ESC	Escape
FF	Form feed	FS	File separator
CR	Carriage return	GS	Group separator
SO	Shift out	RS	Record separator
SI	Shift in	US	Unit separator
SP	Space	DEL	Delete

Error-Detecting Code

- Error-Detecting Code
  - To detect errors in data communication and processing, an eighth bit is added to the ASCII character to indicate its parity.
  - Redundancy (e.g. extra information), in the form of extra bits, can be incorporated into binary code words to detect and correct errors.
  - A simple form of redundancy is parity, an extra bit appended onto the code word to make the number of 1's odd or even. Parity can detect all single-bit errors and some multiple-bit errors.
  - A code word has even parity if the number of 1's in the code word is even.
  - A code word has odd parity if the number of 1's in the code word is odd.

Error-Detecting Code

- **Example:**
  - Consider the following two characters and their even and odd parity:

	Even parity	Odd parity
ASCII A = 1000001	01000001	11000001
ASCII B = 1010100	11010100	01010100

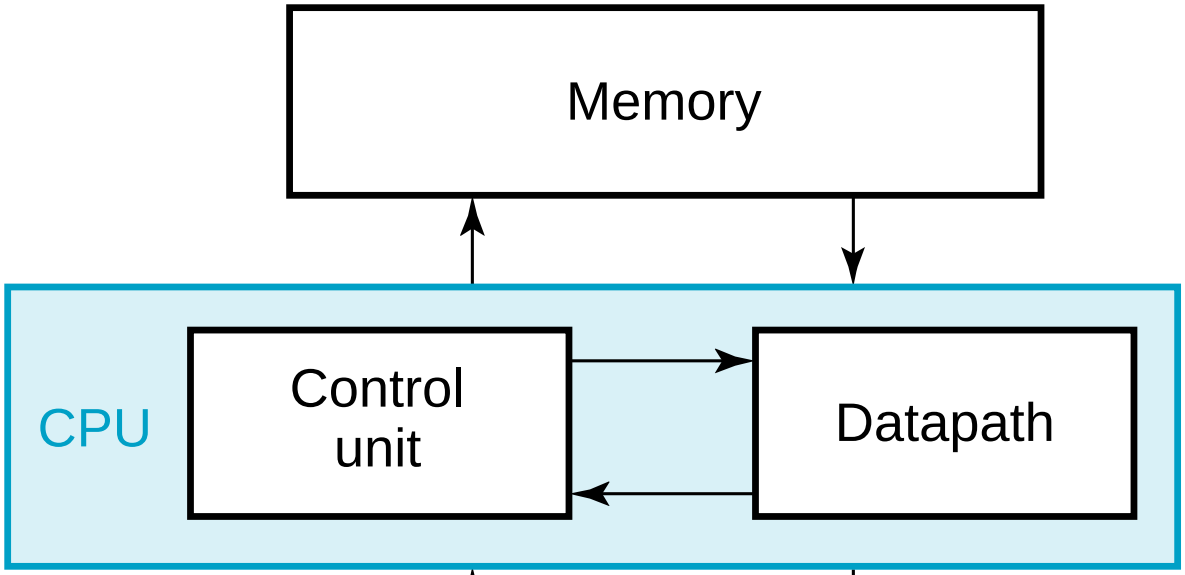
Binary Storage and Registers

- Registers
  - A **binary cell** is a device that *possesses two stable states* and is capable of storing one of the two states.
  - A **register** is a group of binary cells, where a register with *n* cells can store any discrete quantity of information that contains *n* bits.

n cells → 2^n possible states

- A binary cell
  - Two stable state
  - Store one bit of information
  - Examples: flip-flop circuits, capacitor
- A register
  - A group of binary cells
  - AX in x86 CPU
- Register Transfer
  - A transfer of the information stored in one register to another.
  - One of the major operations in digital system.

A Digital Computer Example

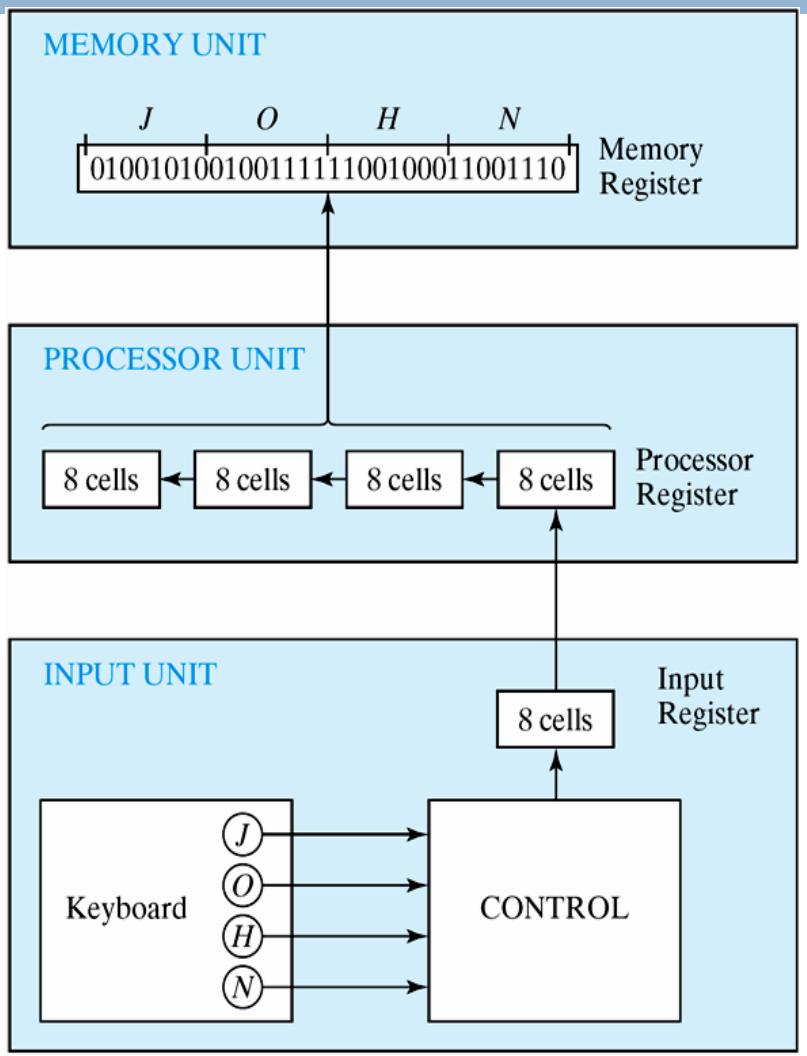


Inputs: Keyboard, mouse, modem, microphone

Outputs: CRT, LCD, modem, speakers

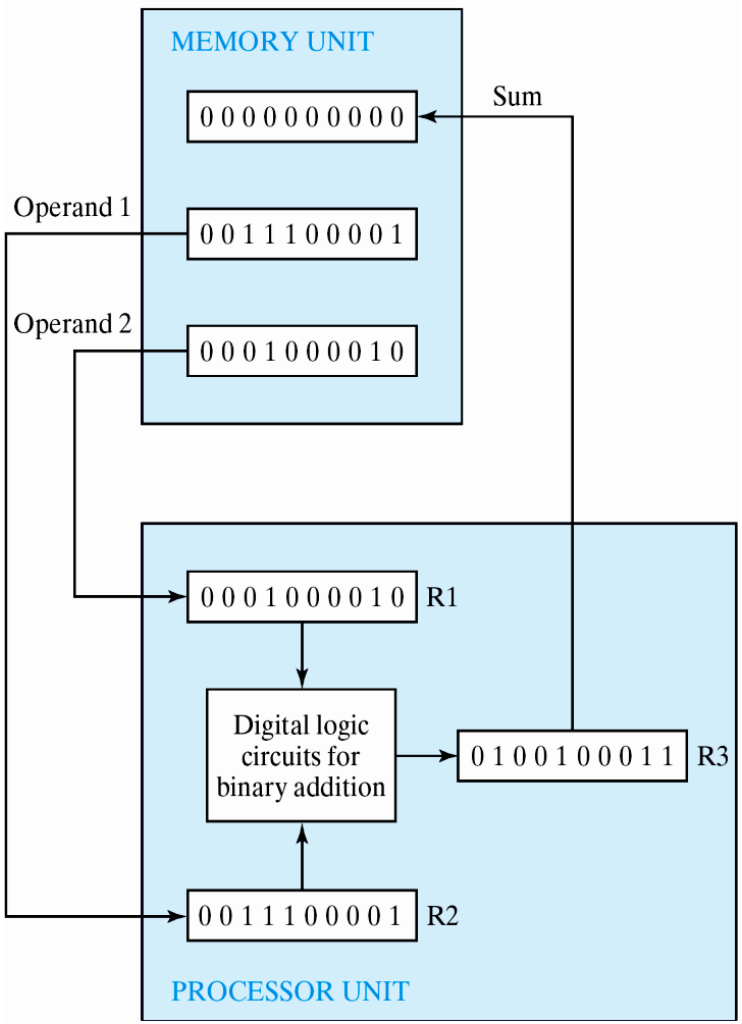
Synchronous or Asynchronous?

Transfer of Information



Transfer of information among register

Transfer of Information



Example of binary information processing

- The other major component of a digital system
  - Circuit elements to manipulate individual bits of information
  - Load-store machine

```
LD      R1;
LD      R2;
ADD     R3, R2, R1;
SD      R3;
```

Binary Logic

- Definition of Binary Logic
  - **Binary logic** consists of *binary variables* & a *set of logical operations*.
  - The *variables* are designated by letters of the alphabet, such as *A*, *B*, *C*, *x*, *y*, *z*, etc, with each variable having two distinct possible values: 1 and 0,
  - Three basic *logical operations*: AND, OR, and NOT.

Binary Logic

1. AND: This operation is represented by a dot or by the absence of an operator. For example, $x \cdot y = z$ or $xy = z$ is read "x AND y is equal to z," The logical operation AND is interpreted to mean that $z = 1$ if only $x = 1$ and $y = 1$; otherwise $z = 0$. (Remember that x, y, and z are binary variables and can be equal either to 1 or 0, and nothing else.)
2. OR: This operation is represented by a plus sign. For example, $x + y = z$ is read "x OR y is equal to z," meaning that $z = 1$ if $x = 1$ or $y = 1$ or if both $x = 1$ and $y = 1$. If both $x = 0$ and $y = 0$, then $z = 0$.
3. NOT: This operation is represented by a prime (sometimes by an overbar). For example, $x' = z$ (or $\bar{x} = z$) is read "not x is equal to z," meaning that z is what z is not. In other words, if $x = 1$, then $z = 0$, but if $x = 0$, then $z = 1$, The NOT operation is also referred to as the complement operation, since it changes a 1 to 0 and a 0 to 1.

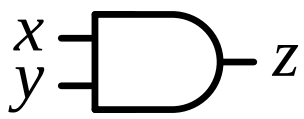
Binary Logic

- Truth Tables, Boolean Expressions, and Logic Gates

AND

x	*y*	*z*
0	0	0
0	1	0
1	0	0
1	1	1

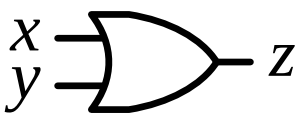
z = x • y = x y



OR

x	*y*	*z*
0	0	0
0	1	1
1	0	1
1	1	1

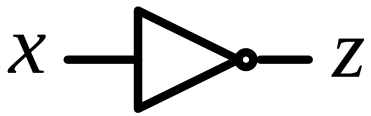
z = x + y



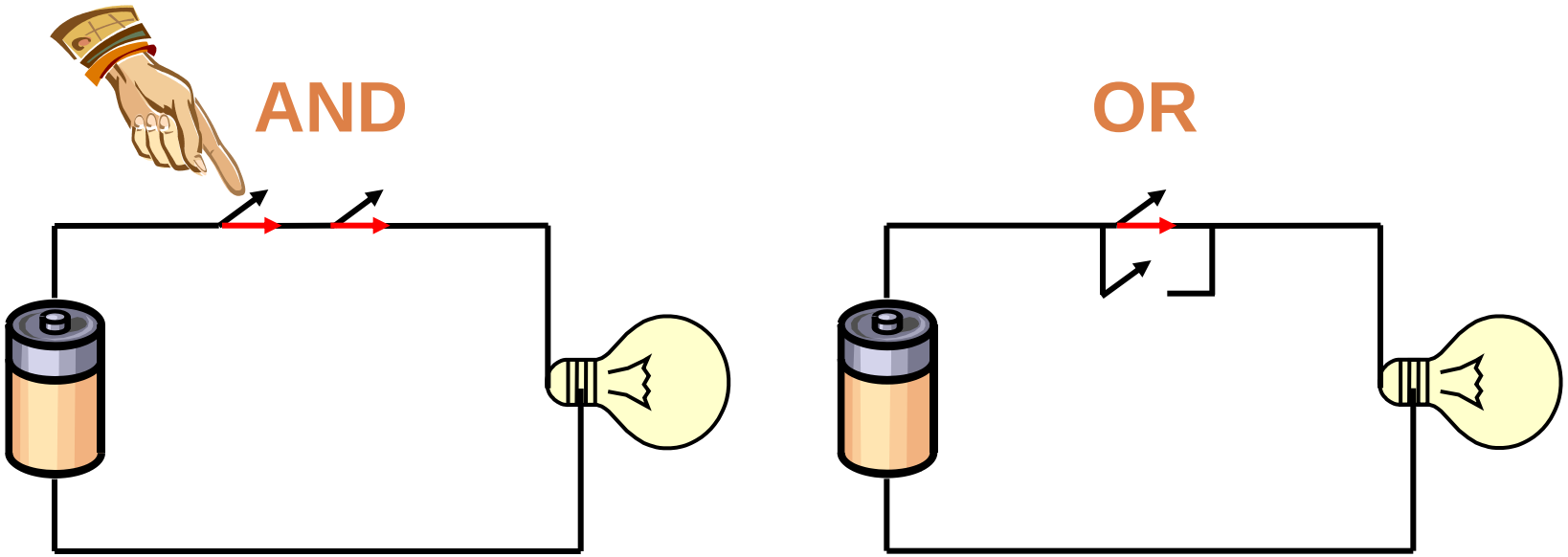
NOT

x	*z*
0	1
1	0

z = $\overline{x}$ = x'

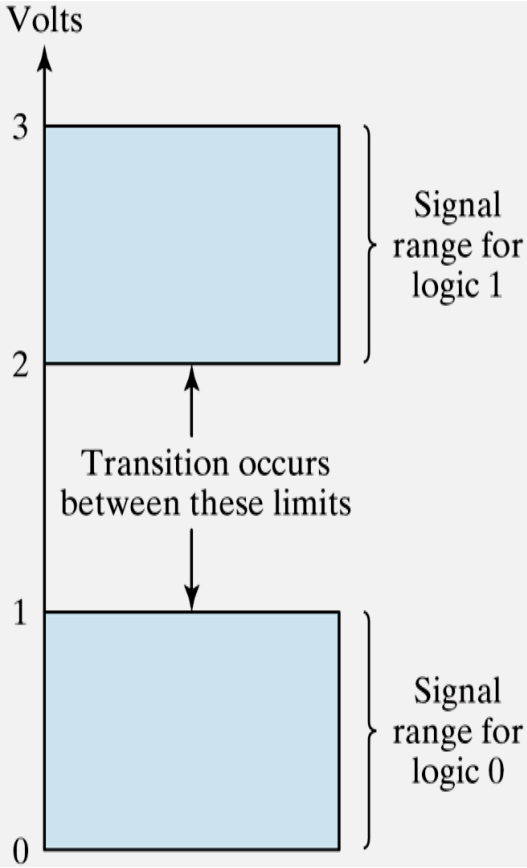


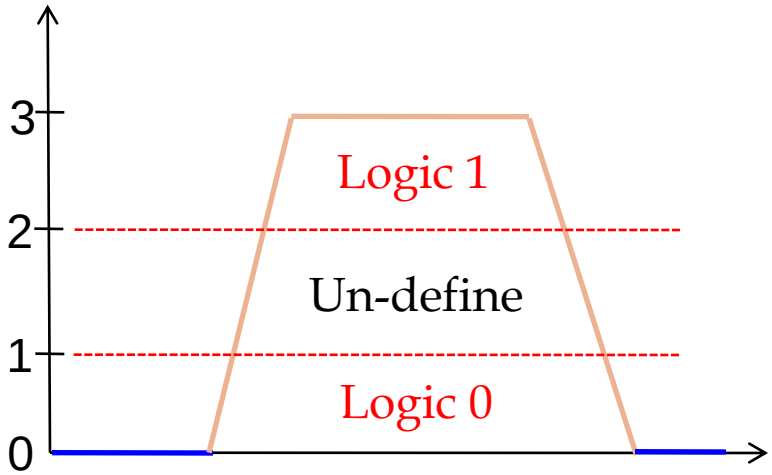
Switching Circuits



Binary Logic

- Logic gates
  - Example of binary signals

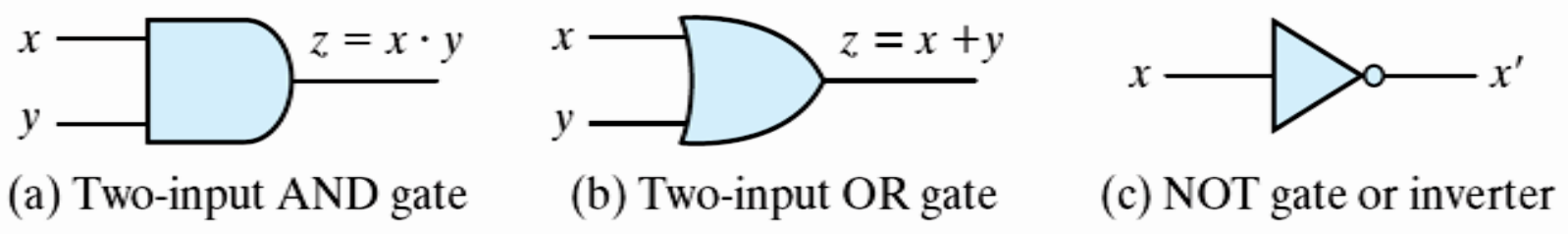




Example of binary signals

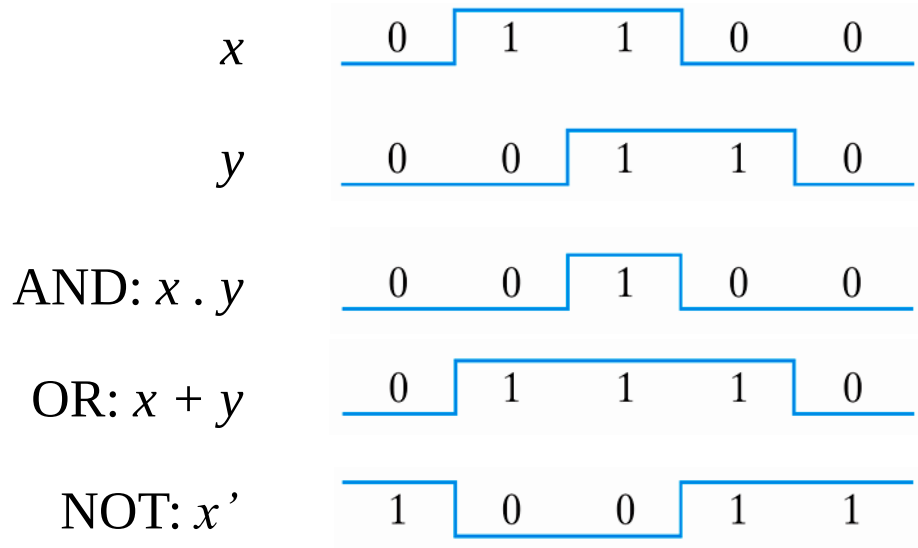
Binary Logic

- Logic gates
  - Graphic Symbols and Input-Output Signals for Logic gates:



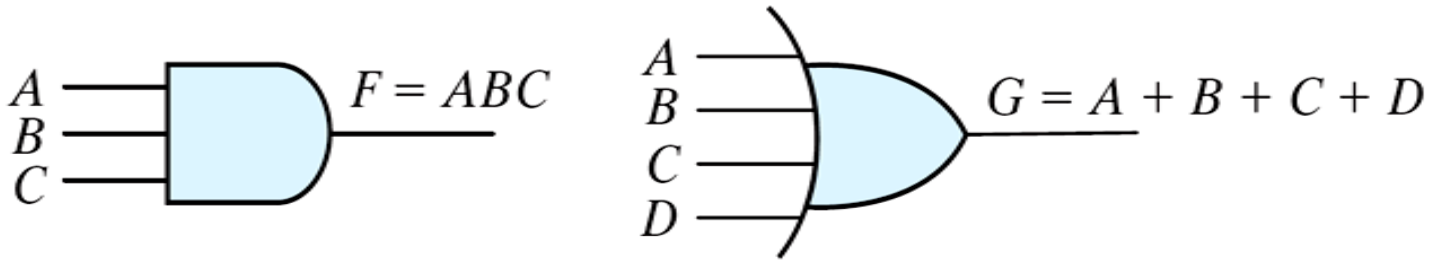
(a) Two-input AND gate (b) Two-input OR gate (c) NOT gate or inverter

- Example:



Binary Logic

- Logic gates
  - Graphic Symbols and Input-Output Signals for Logic gates:



(a) Three-input AND gate (b) Four-input OR gate

Gates with multiple inputs